



Guangdong Technion

Israel Institute of Technology

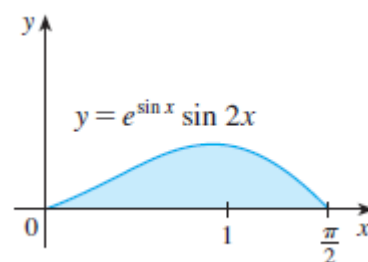
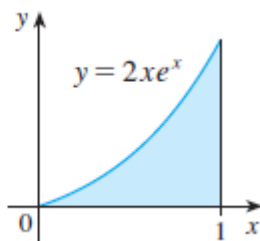
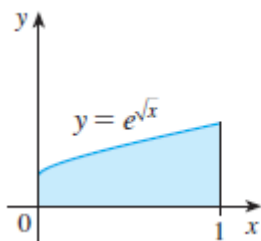
广东以色列理工学院

Homework 6 - Deadline December 13th

Exercise 1. 20 points. Find all the differentiable functions $f : (0, +\infty) \rightarrow (0, +\infty)$ such that $f(e/3) = 1$ and

$$\left[x + x \ln^2(3x) \right] f^2(x) f'(x) = f^3(x)$$

Exercise 2. 20 points. Which of these areas are equal? Why? Make all the computations that you need to justify your answer.



Exercise 3. a) 15 points. Find the area of the region enclosed by the curves

$$f(x) = \frac{x}{10} \text{ and } g(x) = \frac{x}{\sqrt{x^2 + 36}}$$

b) 20 points. Find the volume of the solid obtained by the rotation of the region of part a) around the x -axis.

Exercise 4. a) 25 points. Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function. Prove the identity:

$$\int_0^\pi x f(\sin(x)) dx = \frac{\pi}{2} \int_0^\pi f(\sin(x)) dx$$

b) Use part a) to compute

$$\int_0^\pi \frac{x \sin(x)}{1 + \cos^2(x)} dx$$